# Unit 7 Quiz: Probability Distributions

Name _____

Show your work.

## 7.1 - Probability Distributions

1. **I check if a probability distribution is valid, and I draw it as a graph.**

   (a) Is this table a valid probability distribution? Explain.

| $x$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $P(x)$ | 0.2 | 0.5 | 0.4 | $-0.1$ |

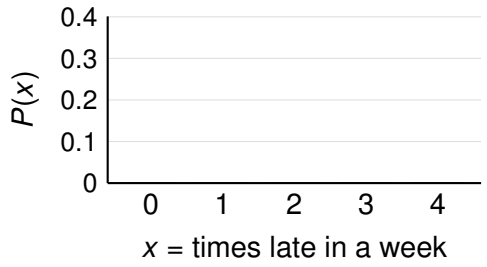
Let $X$ = the number of times the school bus is late in one week.

| $x$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $P(x)$ | 0.35 | 0.30 | ? | 0.10 | 0.05 |

   (b) Find the missing probability, $P(2)$.

   (c) Find $P(X \geq 2)$.

   (d) Draw a bar graph of the distribution.




2. **I explain what a random variable is, and I make theoretical and empirical distributions.**

   A box has 10 cards. 5 cards show the number 1. 3 cards show 2. 2 cards show 5. You draw one card without looking. Let $X$ = the number on the card.

   (a) Why is $X$ a random variable?

   (b) Make the **theoretical** probability distribution of $X$.

| $x$ | 1 | 2 | 5 |
|---|---|---|---|
| $P(x)$ | | | |

   Ms. Diaz's class drew a card 50 times. They put the card back each time. They got a 1 on 22 draws, a 2 on 18 draws, and a 5 on 10 draws.

   (c) Make the **empirical** probability distribution of $X$.

| $x$ | 1 | 2 | 5 |
|---|---|---|---|
| $P(x)$ | | | |

   (d) Why are the two distributions not the same?

## 7.2 - Expected Value

3. **I find the expected value of a random variable, and I explain what it means.**

A basketball team scores 0, 1, 2, or 3 points each time it has the ball. Let $X$ = the number of points it scores on one turn with the ball.

| $x$ (points) | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $P(x)$ | 0.45 | 0.05 | 0.35 | 0.15 |
| $x \cdot P(x)$ | | | | |

(a) Fill in the last row of the table.

(b) Find $E(X)$.

(c) Which statement best describes the meaning of $E(X)$? Circle the letter.

A The team scores exactly $E(X)$ points each time it has the ball.
B Over many turns with the ball, the team scores about $E(X)$ points per turn, on average.
C $E(X)$ is the number of points the team scores most often.
D Over many turns with the ball, the team scores $E(X)$ points in total.

(d) The team has the ball 60 times in a game. About how many points do you expect it to score?

4. **I decide if a method of choosing is fair, and I decide if a game is fair.**

Lin, Omar, and Sara need to choose who presents first.

(a) Which method is fair? Circle the letter.

A Roll a number cube. 1 or 2: Lin. 3: Omar. 4, 5, or 6: Sara.
B Roll a number cube. 1 or 6: Lin. 2 or 5: Omar. 3 or 4: Sara.
C Flip a coin 3 times. 3 heads: Lin. 3 tails: Omar. Anything else: Sara.
D Spin a spinner with 5 equal parts. 2 parts: Lin. 2 parts: Omar. 1 part: Sara.

(b) Explain your choice.

At a school fair game booth, you roll a number cube. You win $4 if you roll a 1 or a 2. You pay $3 for any other number.

(c) Find the expected value of what you win.

(d) Is the game fair? Explain.

(e) How much should you win for a 1 or a 2 to make the game fair?

## 7.3 - Expected Payoff

5. **I find the net gain for each outcome, and I find the expected payoff of a lottery, game, or plan.**

   The soccer team sells 100 raffle tickets for $5 each. One ticket wins $200. Five tickets win $20 each. The rest win nothing.

   (a) Fill in the table.

| Outcome | Net gain $x$ | $P(x)$ | $x \cdot P(x)$ |
|---|---|---|---|
| Win $200 | | | |
| Win $20 | | | |
| Win nothing | | | |

   (b) Find the expected payoff.

   (c) What does the expected payoff mean?

   (d) The team sells all 100 tickets. How much money does the team raise?

6. **I use expected value, and risk, to make a decision.**

   A tablet screen costs $180 to fix. The chance that a screen breaks in one year is 0.15. A one-year protection plan costs $35.

   (a) Find the expected cost to fix the screen with no plan.

   (b) Which is cheaper on average: buying the plan or not? By how much?

   (c) Give a reason someone might still buy the plan.

   (d) The company sells 2,000 of these plans. About how much money does it expect to make?